



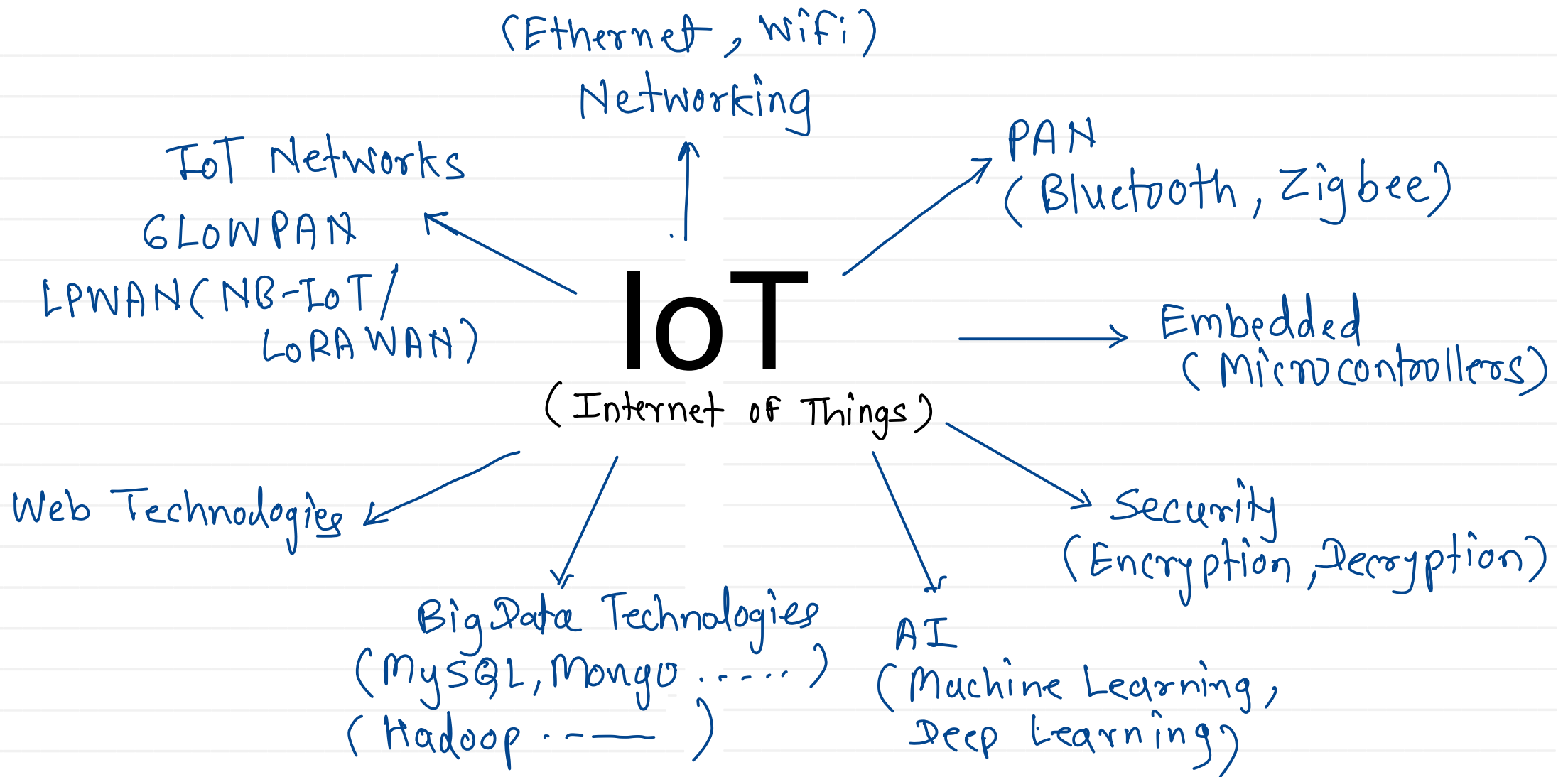
Sunbeam Institute of Information Technology

Pune and Karad

Module – Internet of Things (IoT)

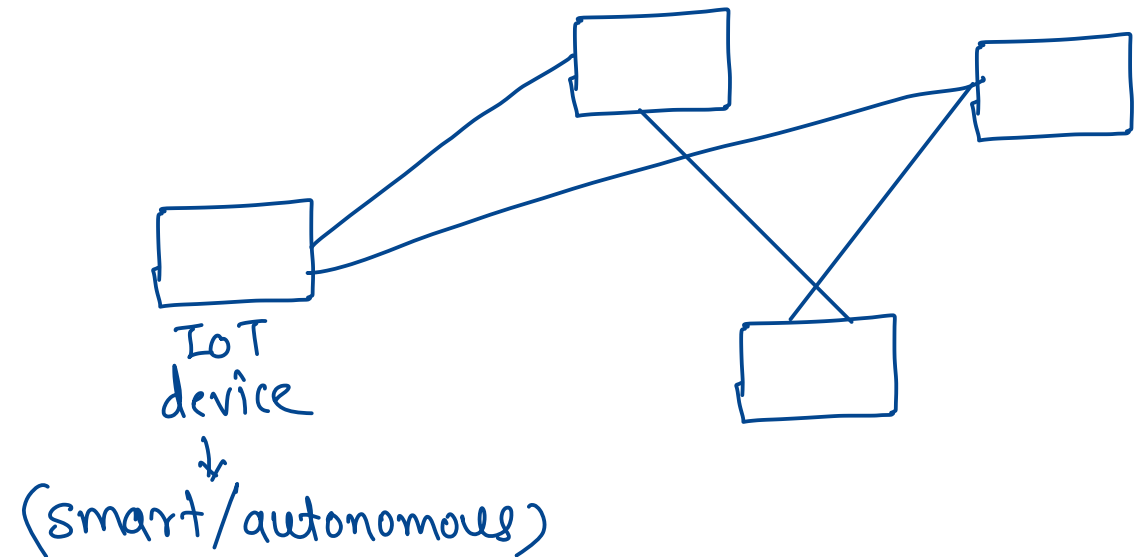
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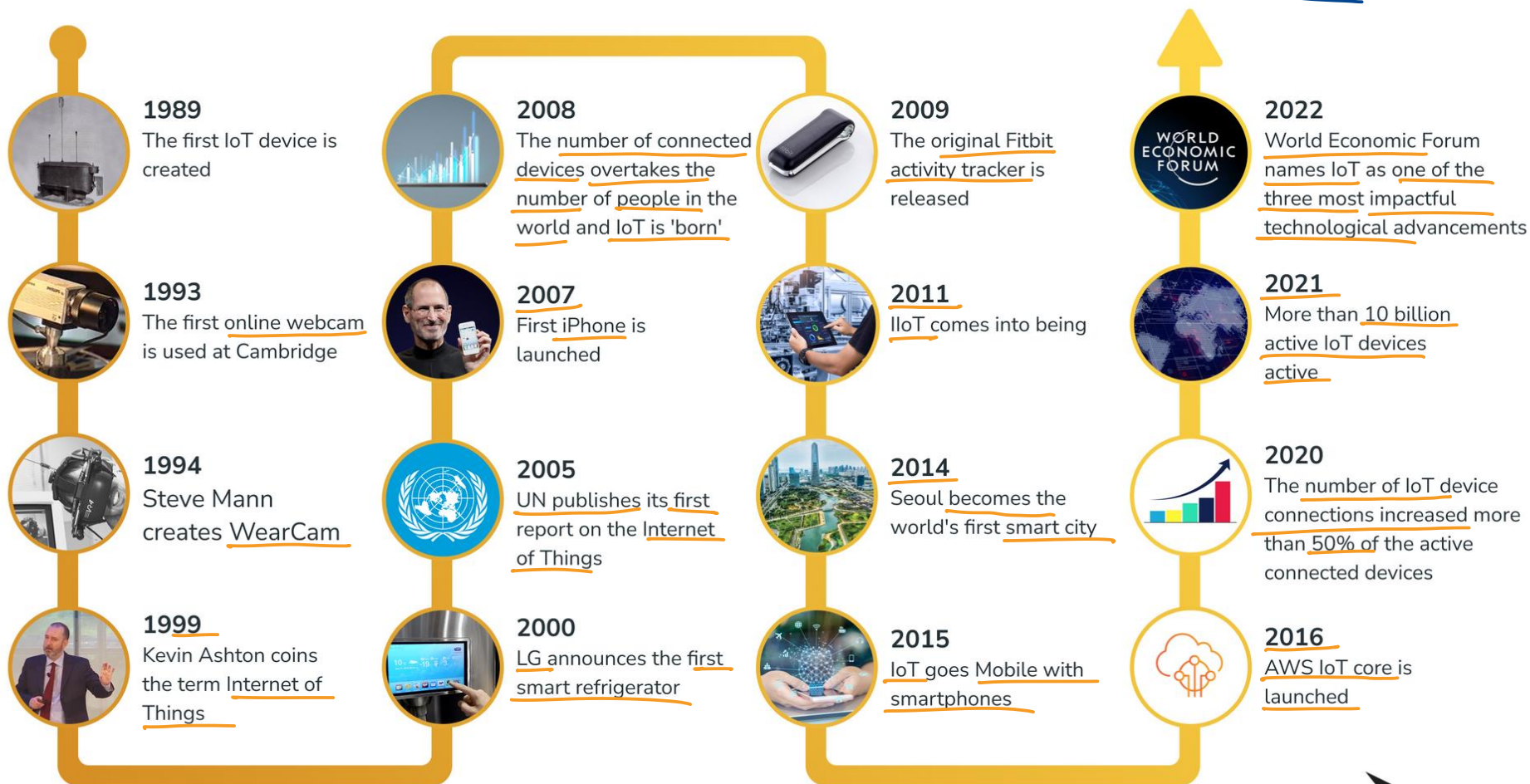
Internet of Things (IoT)

- **Internet of things (IoT)** describes devices with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communication networks.
- **Internet of Things (IoT)** refers to a network of physical devices, vehicles, appliances, and other physical objects that are embedded with sensors, software, and network connectivity, allowing them to collect and share data.



Internet of Things (IoT) – Evolution and Growth

- Kevin Ashton, the co-founder of the Auto-ID Labs at MIT, coined the term 'Internet of Things' in 1999.



Benefits of Internet of Things (IoT)

- **Improved efficiency**

- IoT devices to automate and optimize processes, businesses can improve efficiency and productivity.

- **Data-driven decision-making**

- IoT devices generate vast data that can be used to make better business decisions and new business models.
- By analyzing this data, businesses can gain insights into customer behavior, market trends, and operational performance which helps to take decisions about strategy, product development, and resource allocation.

- **Cost-savings**

- By reducing manual processes and automating repetitive tasks, IoT can help businesses reduce costs and improve profitability.

- **Enhanced customer experience**

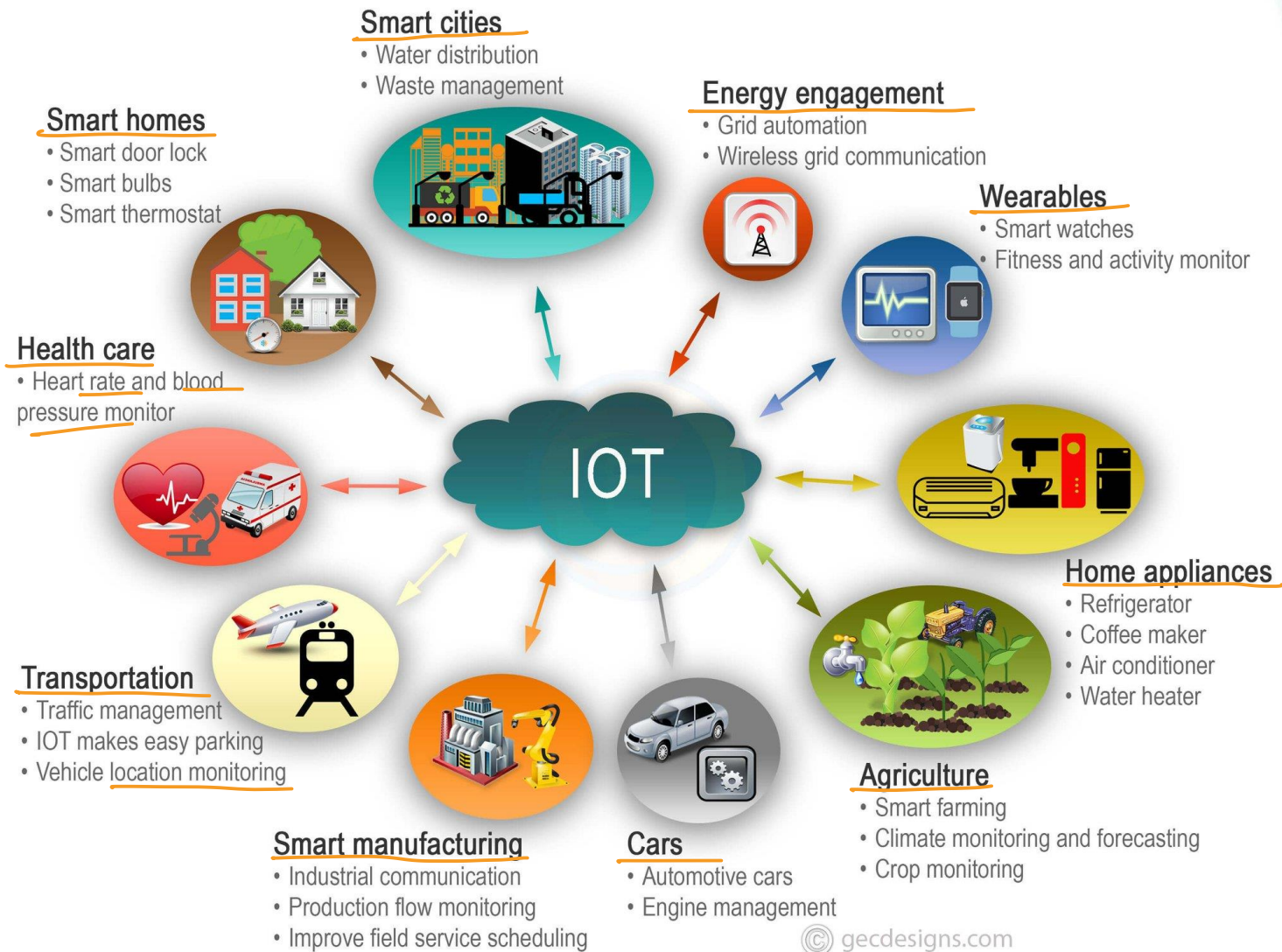
- By using IoT technology to gather data about customer behavior, businesses can create more personalized and engaging experiences for their customers.

Challenges in Internet of Things (IoT)

- **Security and privacy risks**
 - Many IoT devices are vulnerable to hackers and other cyber threats, which can compromise the security and privacy of sensitive data.
 - IoT devices can also collect vast amounts of personal data, raising concerns about privacy and data protection.
- **Interoperability issues**
 - IoT devices from different manufacturers often use different standards and protocols, making it difficult for them to perform “machine to machine” communication.
- **Data overload**
 - IoT devices generate vast data, which can overwhelm businesses that are not prepared to handle it.
 - Analyzing this data and extracting meaningful insights can be a significant challenge
- **Cost and complexity**
 - Implementing an IoT system can be costly and complex, requiring significant investments in hardware, software, and infrastructure.
 - Managing and maintaining an IoT system can also be challenging, requiring specialized skills and expertise.
- **Regulatory and legal challenges**
 - Businesses need to comply with various data protection, privacy and cybersecurity regulations, which can vary from country to country.

Internet of Things (IoT) Applications

- Healthcare
- Manufacturing
- Retail
- Agriculture
- Transportation
- Consumers
- Home automation
- Industrial
- Infrastructure
- Energy management
- Environmental monitoring



- **Growth**

- number of IoT devices is expected to grow rapidly, with estimation of tens of billion IoT devices in use over the next few years

- **Edge computing**

- important for IoT, as it allows data to be processed and analyzed closer to the source of the data, rather than in a centralized data center.
- This can improve response times, reduce latency and reduce the amount of data that needs to be transferred over IoT networks.

- **Artificial intelligence and machine learning**

- used to analyze vast amounts of data that is generated by IoT devices and extract meaningful insights.
- This can help businesses make more informed decisions and optimize their operations.

- **Blockchain**

- explored as a way to improve security and privacy in the IoT.
- Blockchain can be used to create secure, decentralized networks for IoT devices, which can minimize data security vulnerabilities.

- **Sustainability**

- IoT can be used to optimize energy usage, reduce waste and improve sustainability across a range of industries.

Several technologies come together to make IoT possible.

- **Sensors and actuators**

- Sensors are devices that can detect changes in the environment, such as temperature, humidity, light, motion, or pressure.
- Actuators are devices that can cause physical changes in the environment, such as opening or closing a valve or turning on a motor.
- Automation is possible when sensors and actuators work to resolve issues without human intervention.

- **Connectivity technologies**

- To transmit IoT data from sensors and actuators to the cloud, IoT devices need to be connected to the internet.
- There are several connectivity technologies that are used in IoT, including wifi, Bluetooth, cellular, Zigbee, and LoRaWAN.

- **Cloud computing**

- platforms provide the infrastructure and tools that are needed to store and analyze this data, as well as to build and deploy IoT applications.

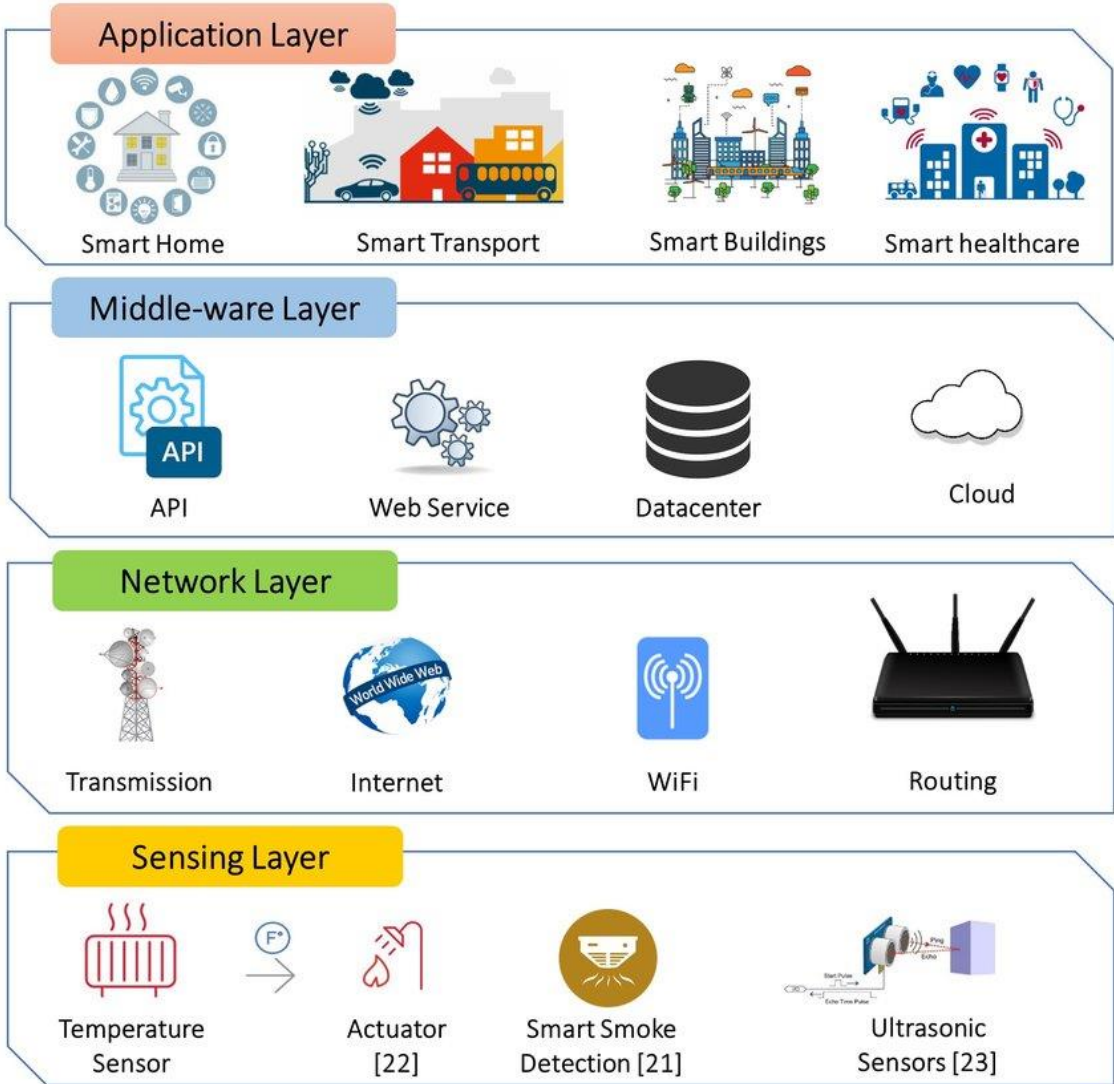
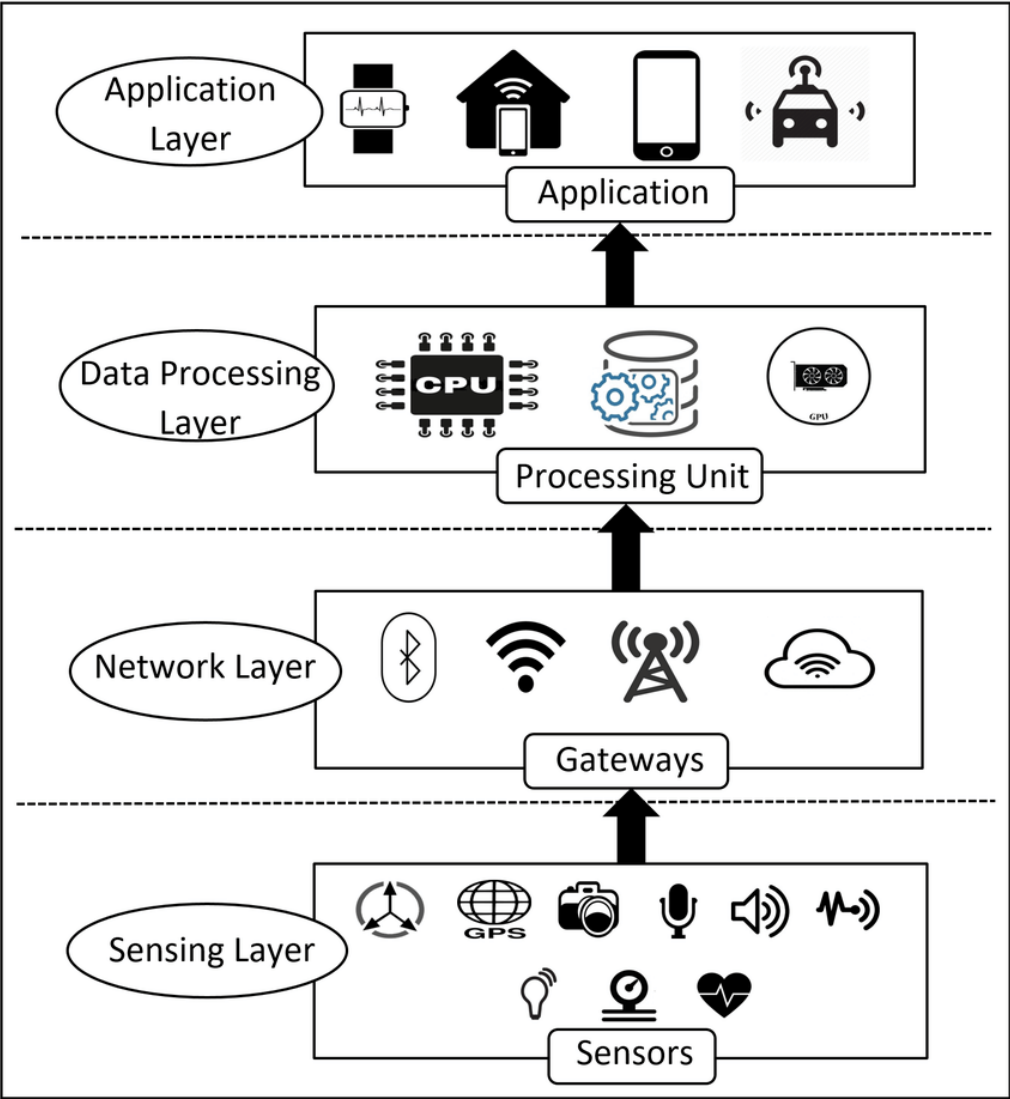
Several technologies come together to make IoT possible.

- **Big data analytics**
 - data generated by IoT devices need to use advanced analytics tools to extract insights and identify patterns.
 - These tools can include machine learning algorithms, data visualization tools and predictive analytics models.
- **Security and privacy technologies**
 - Technologies such as encryption, access controls and intrusion detection systems are used to protect IoT devices

Internet of Things (IoT) Architecture

- **Perception/Sensing Layer**
 - Perception refers to the physical layer, which includes sensors and actuators that are capable of collecting, accepting, and processing data over the network.
 - Sensors and actuators can be connected either wirelessly or via wired connections.
- **Network Layer**
 - This layer contains Data Acquiring Systems (DAS) and Internet/Network gateways.
 - It is necessary to transmit and process the data collected by the sensor devices.
 - This layer allows these devices to connect and communicate with other servers, smart devices, and network devices.
- **Processing Layer**
 - The processing layer is the brain of the IoT ecosystem.
 - Data is analyzed, pre-processed, and stored here before being sent to the data center
 - Data is accessed by software applications that both monitor and manage the data as well as prepare further actions.
- **Application Layer**
 - User interaction takes place at the application layer, which delivers application-specific services to the user.

Internet of Things (IoT) Architecture





Thank you!!!

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